Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Ken VanFoeken Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
9375 N Washington RD	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1945

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X130-X011-XXXX X045-X130-X040-XXXX X045-X130-X053-XXXX X045-X130-X058-XXXX

B. OPERATORS

VanFoeken, Ken

Operator name: VanFoeken, Ken

Telephone no.: (209) 667-2393 (Landline) (209) 658-3912 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
22338 Short RD	Hilmar	CA	95324

This operator is responsible for paying permit fees.

VanFoeken, Kurtis

Operator name: VanFoeken, Kurtis

Telephone no.: (209) 667-2393 (Landline)

Mailing Address Number and Street	City	State	Zip Code
22338 Short RD	Hilmar	CA	95324

C. OWNERS

VanFoeken, Ken

Legal owner name:	VanFoeken, Ken	Telephone no.:	(209) 667-2393	(209) 658-3912
			Landline	Cellular
22338 Short RD	Hilmar		CA	95324
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	558	67	304	119	109	141
Maximum number	560	68	304	119	109	141
Average number	560	67	304	119	109	141
Avg live weight (lbs)	950	1,050	650	400		

Predominant milk cow breed: Jersey

Average milk production: 66 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 19,676.31 *tons per reporting period*

Total nitrogen from manure: 249,520.87 *lbs per reporting period*

After ammonia losses (30% loss applied): 174,664.61 *lbs per reporting period*

Total phosphorus from manure: 40,474.37 *lbs per reporting period*

Total potassium from manure: 105,773.49 *lbs per reporting period*

Total salt from manure: 279,082.65 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 13,147,390 *gallons*

Total nitrogen generated: 45,436.22 *lbs*

Total phosphorus generated: 7,494.25 *lbs*

Total potassium generated: 63,603.70 *lbs*

Total salt generated: 328,840.88 *lbs*

	7,155,000	*gallons applied*
+	5,992,390	*gallons exported*
-	0	*gallons imported*
=	13,147,390	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - NE Well	Ground water
Domestic Well - NW Well	Ground water
Domestic Well - S Well	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/22/2023	Corral solids	2,400.00 *ton*	Dry-weight	15.7		22,900.00	7,700.00	28,200.00		0.00

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
03/03/2023	Process wastewater	985,500.00 *gal*	431.00	299.00	0.00	0.80	73.60	590.00		3,260
07/14/2023	Process wastewater	5,006,890.00 *gal*	385.00	274.00	0.00	0.40	59.70	563.00		2,570

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	92,662.56	31,157.28	114,108.48	0.00
Process wastewater	19,654.10	3,099.70	28,375.70	134,191.25
Total exports for all materials	112,316.66	34,256.98	142,484.18	134,191.25

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Casavans	11	11	2	manure	X045-X130-X053-XXXX
The 20	20	20	2	process wastewater	X045-X130-X040-XXXX
The 25	25	25	2	process wastewater	X045-X130-X058-XXXX
The 4	4	4	2	process wastewater	X045-X130-X011-XXXX
Totals for areas that were used for application	60	60	8		
Totals for areas that were not used for application					
Land application area totals	60	60	8		

B. CROPS AND HARVESTS

Casavans

Field name: Casavans

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	119.50 *ton*	Dry-weight		74.3	30,100.00	5,000.00	45,000.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.86	168.08	27.92	251.28	558.39

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	305.60 *ton*	Dry-weight		62.6	11,400.00	2,500.00	7,900.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.78	236.90	51.95	164.17	914.36

The 20

Field name: The 20

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	181.20 *ton*	Dry-weight		55.4	22,400.00	4,800.00	37,100.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.06	181.03	38.79	299.82	1,002.11

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	518.60 *ton*	Dry-weight		64.8	15,300.00	3,300.00	23,500.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.93	279.30	60.24	428.99	1,369.10

The 25

Field name: The 25

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	251.40 *ton*	Dry-weight		58.4	24,200.00	4,800.00	41,300.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.06	202.47	40.16	345.54	853.39

The 25

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	653.20 *ton*	Dry-weight		67.0	15,900.00	3,300.00	20,700.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.13	274.19	56.91	356.96	1,327.82

The 4

Field name: The 4

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 4 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	40.00 *ton*	Dry-weight		55.4	22,400.00	4,800.00	37,100.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.00	199.81	42.82	330.93	1,106.08

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 4 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	100.30 *ton*	Dry-weight		64.8	15,300.00	3,300.00	23,500.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.08	270.09	58.25	414.84	1,323.96

NUTRIENT BUDGET

A. LAND APPLICATIONS

Casavans - 11/05/2022: Oats, silage-soft dough

Field name: Casavans

Crop: Oats, silage-soft dough — Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	210.60	70.81	259.34	0.00	60.00 *ton*
Application event totals		210.60	70.81	259.34	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	4.09	0.00	0.00	101.63	1,717,425.00 *gal*
Application event totals		4.09	0.00	0.00	101.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	4.33	0.00	0.00	107.60	1,818,450.00 *gal*
Application event totals		4.33	0.00	0.00	107.60	

Casavans - 05/14/2023: Corn, silage

Field name: Casavans

Crop: Corn, silage — Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	280.80	94.42	345.78	0.00	80.00 *ton*
Application event totals		280.80	94.42	345.78	0.00	

Casavans - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.85	0.00	0.00	95.65	1,616,400.00 *gal*
Application event totals		3.85	0.00	0.00	95.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.61	0.00	0.00	89.67	1,515,375.00 *gal*
Application event totals		3.61	0.00	0.00	89.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	2.89	0.00	0.00	71.74	1,212,300.00 *gal*
Application event totals		2.89	0.00	0.00	71.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.13	0.00	0.00	77.71	1,313,325.00 *gal*
Application event totals		3.13	0.00	0.00	77.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	2.89	0.00	0.00	71.74	1,212,300.00 *gal*
Application event totals		2.89	0.00	0.00	71.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.37	0.00	0.00	83.69	1,414,350.00 *gal*
Application event totals		3.37	0.00	0.00	83.69	

Casavans - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.13	0.00	0.00	77.71	1,313,325.00 *gal*
Application event totals		3.13	0.00	0.00	77.71	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.37	0.00	0.00	83.69	1,414,350.00 *gal*
Application event totals		3.37	0.00	0.00	83.69	

The 20 - 11/05/2022: Oats, silage-soft dough

Field name: The 20

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	105.40	17.97	144.01	795.74	585,000.00 *gal*
Irrigation Water - Canal	Surface water	3.84	0.00	0.00	95.35	2,929,725.00 *gal*
Application event totals		109.24	17.97	144.01	891.09	

03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.05	17.73	142.17	785.54	577,500.00 *gal*
Irrigation Water - Canal	Surface water	4.37	0.00	0.00	108.50	3,333,825.00 *gal*
Application event totals		108.42	17.73	142.17	894.04	

The 20 - 05/14/2023: Corn, silage

The 20 - 05/14/2023: Corn, silage

Field name: The 20

Crop: Corn, silage

Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.31	0.00	0.00	82.20	2,525,625.00 *gal*
Application event totals		3.31	0.00	0.00	82.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.15	19.12	153.25	846.75	622,500.00 *gal*
Irrigation Water - Canal	Surface water	3.18	0.00	0.00	78.91	2,424,600.00 *gal*
Application event totals		115.33	19.12	153.25	925.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.44	0.00	0.00	85.49	2,626,650.00 *gal*
Application event totals		3.44	0.00	0.00	85.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	109.45	18.66	149.55	826.34	607,500.00 *gal*
Irrigation Water - Canal	Surface water	3.57	0.00	0.00	88.77	2,727,675.00 *gal*
Application event totals		113.03	18.66	149.55	915.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.31	0.00	0.00	82.20	2,525,625.00 *gal*
Application event totals		3.31	0.00	0.00	82.20	

The 20 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.44	0.00	0.00	85.49	2,626,650.00 *gal*
Application event totals		3.44	0.00	0.00	85.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.29	16.58	132.94	734.53	540,000.00 *gal*
Irrigation Water - Canal	Surface water	3.31	0.00	0.00	82.20	2,525,625.00 *gal*
Application event totals		100.60	16.58	132.94	816.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.18	0.00	0.00	78.91	2,424,600.00 *gal*
Application event totals		3.18	0.00	0.00	78.91	

The 25 - 11/05/2022: Oats, silage-soft dough

Field name: The 25

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	115.67	19.72	158.05	873.27	802,500.00 *gal*
Irrigation Water - Canal	Surface water	4.66	0.00	0.00	115.73	4,445,100.00 *gal*
Application event totals		120.33	19.72	158.05	989.01	

The 25 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.05	16.03	128.50	710.04	652,500.00 *gal*
Irrigation Water - Canal	Surface water	4.45	0.00	0.00	110.47	4,243,050.00 *gal*
Application event totals		98.50	16.03	128.50	820.52	

The 25 - 05/14/2023: Corn, silage

Field name: The 25

Crop: Corn, silage

Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.07	0.00	0.00	76.28	2,929,725.00 *gal*
Application event totals		3.07	0.00	0.00	76.28	

05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.02	18.24	146.23	807.98	742,500.00 *gal*
Irrigation Water - Canal	Surface water	3.07	0.00	0.00	76.28	2,929,725.00 *gal*
Application event totals		110.09	18.24	146.23	884.26	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.18	0.00	0.00	78.91	3,030,750.00 *gal*
Application event totals		3.18	0.00	0.00	78.91	

The 25 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.86	17.87	143.28	791.66	727,500.00 *gal*
Irrigation Water - Canal	Surface water	3.39	0.00	0.00	84.17	3,232,800.00 *gal*
Application event totals		108.25	17.87	143.28	875.83	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.07	0.00	0.00	76.28	2,929,725.00 *gal*
Application event totals		3.07	0.00	0.00	76.28	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.07	0.00	0.00	76.28	2,929,725.00 *gal*
Application event totals		3.07	0.00	0.00	76.28	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.70	17.50	140.32	775.33	712,500.00 *gal*
Irrigation Water - Canal	Surface water	3.18	0.00	0.00	78.91	3,030,750.00 *gal*
Application event totals		105.87	17.50	140.32	854.24	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	2.96	0.00	0.00	73.65	2,828,700.00 *gal*
Application event totals		2.96	0.00	0.00	73.65	

The 4 - 11/05/2022: Oats, silage-soft dough

Field name: The 4

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

The 4 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	114.86	19.58	156.94	867.15	127,500.00 *gal*
Irrigation Water - Canal	Surface water	2.65	0.00	0.00	65.76	404,100.00 *gal*
Application event totals		117.50	19.58	156.94	932.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.59	16.12	129.24	714.12	105,000.00 *gal*
Irrigation Water - Canal	Surface water	3.31	0.00	0.00	82.20	505,125.00 *gal*
Application event totals		97.90	16.12	129.24	796.32	

The 4 - 05/14/2023: Corn, silage

Field name: The 4

Crop: Corn, silage

Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	5.29	0.00	0.00	131.52	808,200.00 *gal*
Application event totals		5.29	0.00	0.00	131.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	114.86	19.58	156.94	867.15	127,500.00 *gal*
Irrigation Water - Canal	Surface water	3.97	0.00	0.00	98.64	606,150.00 *gal*
Application event totals		118.83	19.58	156.94	965.79	

The 4 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.31	0.00	0.00	82.20	505,125.00 *gal*
Application event totals		3.31	0.00	0.00	82.20	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.10	18.43	147.71	816.14	120,000.00 *gal*
Irrigation Water - Canal	Surface water	3.31	0.00	0.00	82.20	505,125.00 *gal*
Application event totals		111.41	18.43	147.71	898.34	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.31	0.00	0.00	82.20	505,125.00 *gal*
Application event totals		3.31	0.00	0.00	82.20	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	3.97	0.00	0.00	98.64	606,150.00 *gal*
Application event totals		3.97	0.00	0.00	98.64	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.59	16.12	129.24	714.12	105,000.00 *gal*
Irrigation Water - Canal	Surface water	2.65	0.00	0.00	65.76	404,100.00 *gal*
Application event totals		97.24	16.12	129.24	779.88	

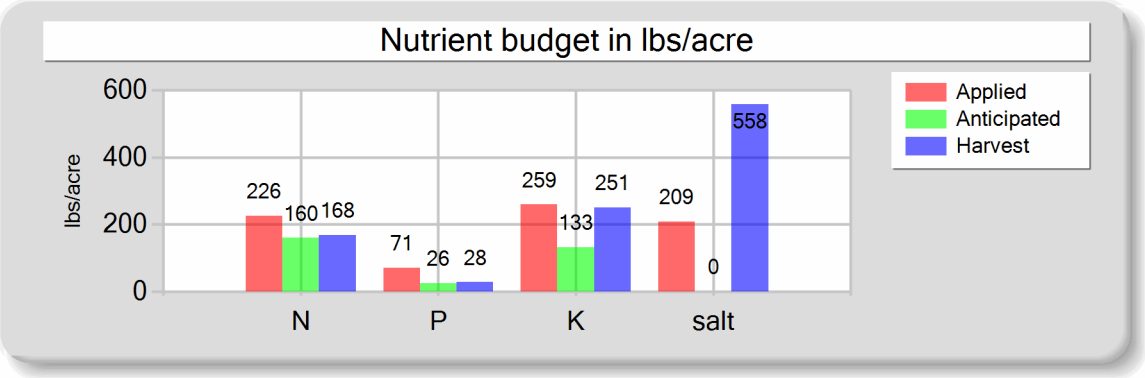
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	2.65	0.00	0.00	65.76	404,100.00 *gal*
Application event totals		2.65	0.00	0.00	65.76	

B. NUTRIENT BUDGET

Casavans - 11/05/2022: Oats, silage-soft dough

Field name: Casavans Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	210.60	70.81	259.34	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.42	0.00	0.00	209.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.02	70.81	259.34	209.23
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	168.08	27.92	251.28	558.39
Nutrient balance	57.94	42.89	8.06	-349.16
Applied to removed ratio	1.34	2.54	1.03	0.37

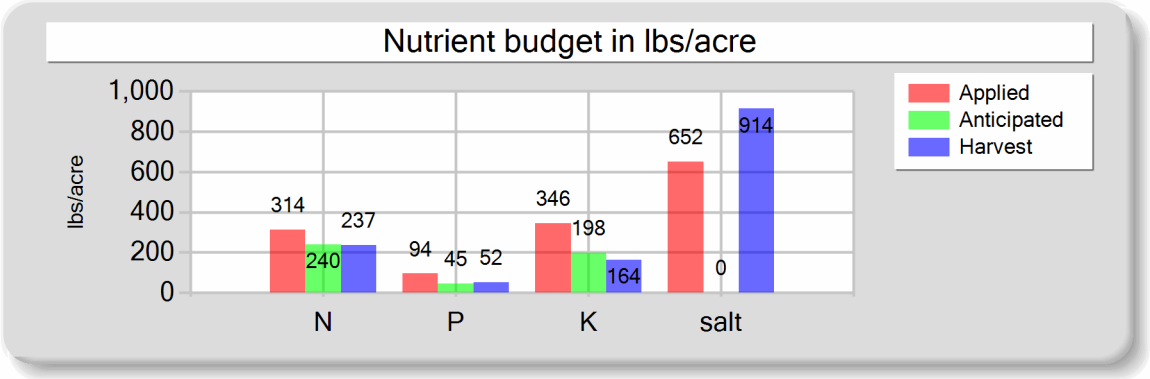
Fresh water applied	
3,535,875.00	*gallons*
130.21	*acre-inches*
11.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Casavans - 05/14/2023: Corn, silage

Field name: Casavans Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	280.80	94.42	345.78	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	26.23	0.00	0.00	651.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.03	94.42	345.78	651.60
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.90	51.95	164.17	914.36
Nutrient balance	77.13	42.46	181.61	-262.75
Applied to removed ratio	1.33	1.82	2.11	0.71

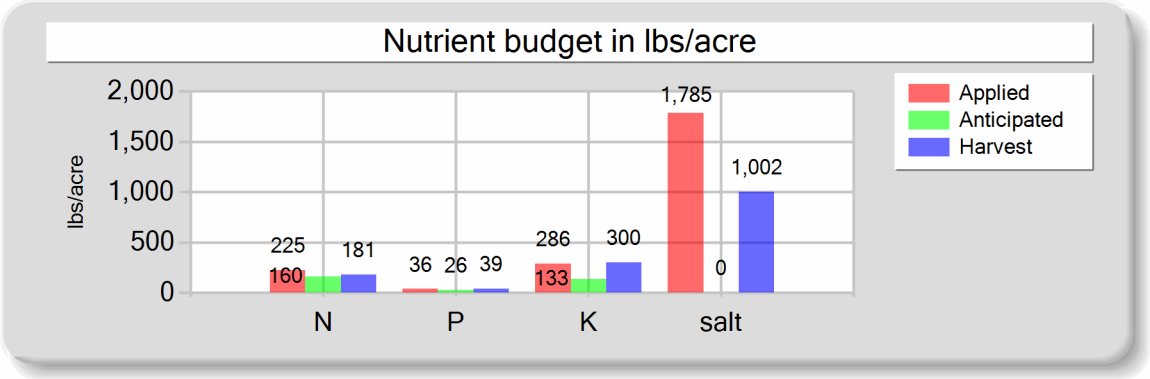
Fresh water applied	
11,011,725.00	*gallons*
405.52	*acre-inches*
36.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 20 - 11/05/2022: Oats, silage-soft dough

Field name: The 20 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.45	35.70	286.18	1,581.27
Fresh water	8.21	0.00	0.00	203.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.65	35.70	286.18	1,785.12
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	181.03	38.79	299.82	1,002.11
Nutrient balance	43.63	-3.09	-13.64	783.02
Applied to removed ratio	1.24	0.92	0.95	1.78

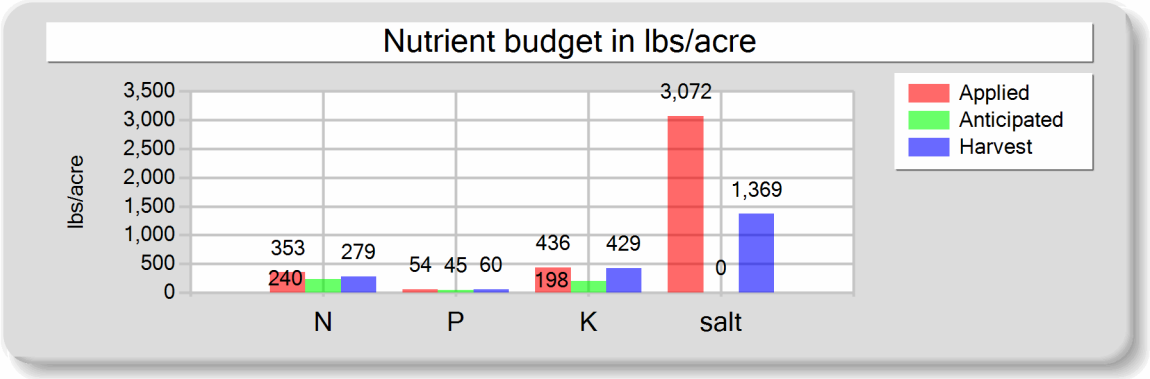
Fresh water applied	
6,263,550.00	*gallons*
230.67	*acre-inches*
11.53	*inches/acre*

Process wastewater applied	
1,162,500.00	*gallons*
42.81	*acre-inches*
2.14	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 20 - 05/14/2023: Corn, silage

Field name: The 20 Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	318.90	54.36	435.73	2,407.62
Fresh water	26.74	0.00	0.00	664.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	352.63	54.36	435.73	3,071.77
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	279.30	60.24	428.99	1,369.10
Nutrient balance	73.34	-5.88	6.75	1,702.67
Applied to removed ratio	1.26	0.90	1.02	2.24

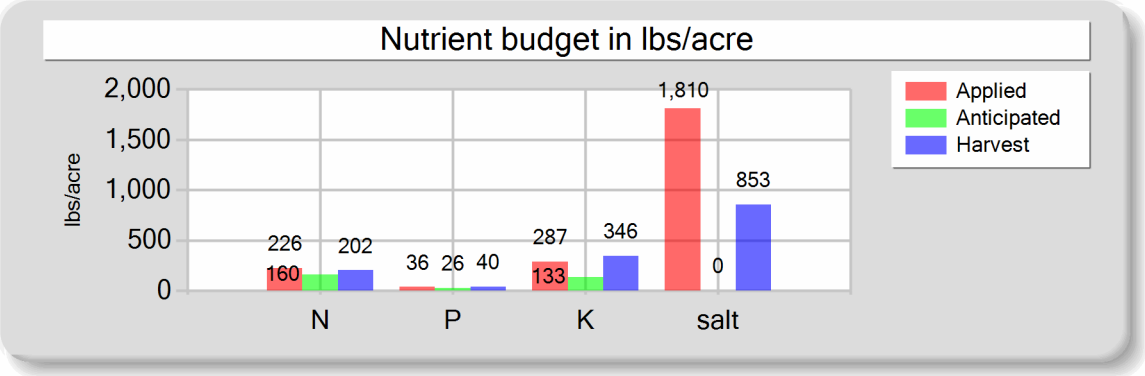
Fresh water applied	
20,407,050.00	*gallons*
751.52	*acre-inches*
37.58	*inches/acre*

Process wastewater applied	
1,770,000.00	*gallons*
65.18	*acre-inches*
3.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 25 - 11/05/2022: Oats, silage-soft dough

Field name: The 25 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.72	35.75	286.55	1,583.31
Fresh water	9.11	0.00	0.00	226.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.82	35.75	286.55	1,809.52
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	202.47	40.16	345.54	853.39
Nutrient balance	23.35	-4.41	-58.99	956.13
Applied to removed ratio	1.12	0.89	0.83	2.12

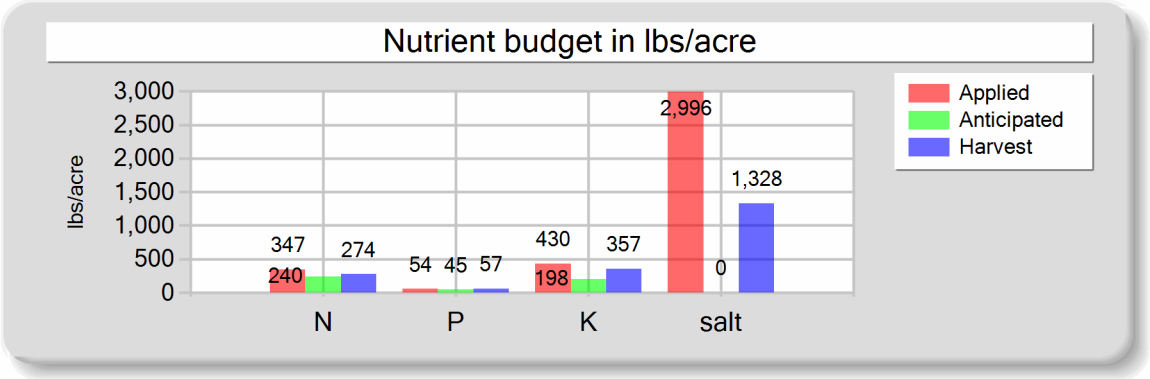
Fresh water applied	
8,688,150.00	*gallons*
319.96	*acre-inches*
12.80	*inches/acre*

Process wastewater applied	
1,455,000.00	*gallons*
53.58	*acre-inches*
2.14	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 25 - 05/14/2023: Corn, silage

Field name: The 25 Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	314.57	53.62	429.83	2,374.97
Fresh water	24.99	0.00	0.00	620.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	346.56	53.62	429.83	2,995.73
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	274.19	56.91	356.96	1,327.82
Nutrient balance	72.38	-3.29	72.87	1,667.90
Applied to removed ratio	1.26	0.94	1.20	2.26

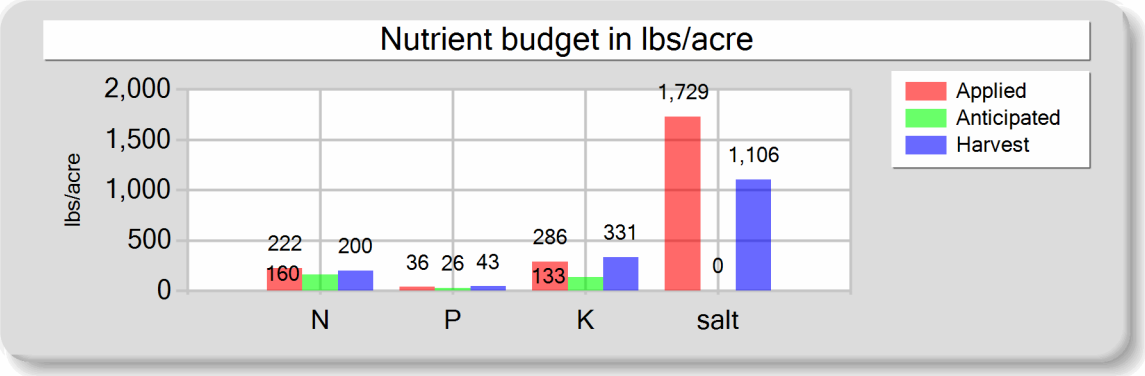
Fresh water applied	
23,841,900.00	*gallons*
878.02	*acre-inches*
35.12	*inches/acre*

Process wastewater applied	
2,182,500.00	*gallons*
80.37	*acre-inches*
3.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 4 - 11/05/2022: Oats, silage-soft dough

Field name: The 4 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.45	35.70	286.18	1,581.27
Fresh water	5.96	0.00	0.00	147.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.40	35.70	286.18	1,729.23
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	199.81	42.82	330.93	1,106.08
Nutrient balance	22.59	-7.12	-44.75	623.15
Applied to removed ratio	1.11	0.83	0.86	1.56

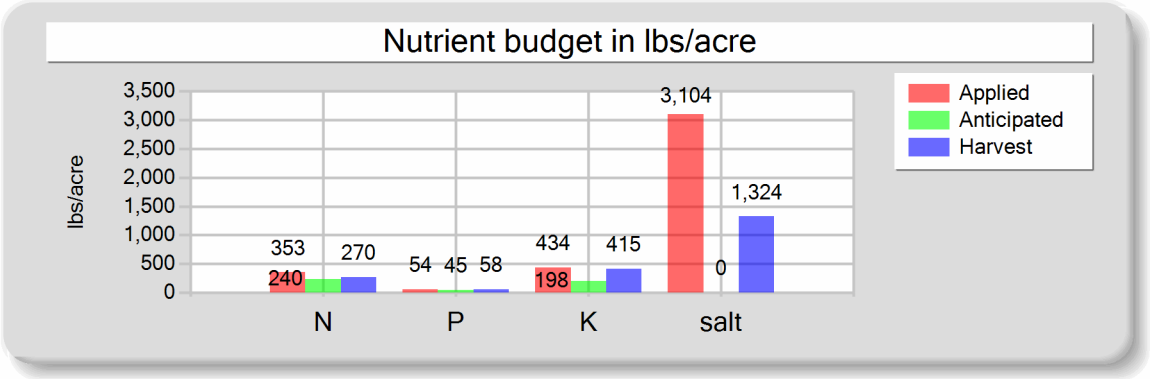
Fresh water applied	
909,225.00	*gallons*
33.48	*acre-inches*
8.37	*inches/acre*

Process wastewater applied	
232,500.00	*gallons*
8.56	*acre-inches*
2.14	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 4 - 05/14/2023: Corn, silage

Field name: The 4 Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	317.55	54.13	433.89	2,397.41
Fresh water	28.46	0.00	0.00	706.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	353.00	54.13	433.89	3,104.31
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	270.09	58.25	414.84	1,323.96
Nutrient balance	82.92	-4.13	19.05	1,780.35
Applied to removed ratio	1.31	0.93	1.05	2.34

Fresh water applied	
4,344,075.00	*gallons*
159.98	*acre-inches*
39.99	*inches/acre*

Process wastewater applied	
352,500.00	*gallons*
12.98	*acre-inches*
3.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/22/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 15.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	7,700.00	28,200.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,600.00	3,100.00	16,200.00							0.00
DL	100.00	100.00	100.00							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 03/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	431.00	299.00	0.00	0.80	73.60	590.00								6,660.00	3,260
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon water

Sample and source description: Lagoon water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	385.00	274.00	0.00	0.40	59.70	563.00								6,150.00	2,570
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	270.00	123.00	0.00	13.20	69.90	240.00								2,790.00	1,670
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	344.00	298.00	0.00	0.10	52.20	817.00								8,120.00	4,450
DL	1.00	0.50	1.00	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - NE Well

Domestic Well - NE Well

NE Well

Sample description: NE Well

Sample date: 11/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	68.00		68.00								1,740.00	
DL	0.10		0.10								10.00	

Domestic Well - NW Well

NW Well

Sample description: NW Well

Sample date: 11/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	63.70		63.70								1,710.00	
DL	0.10		0.10								10.00	

Domestic Well - S Well

S Well

Sample description: S Well

Sample date: 11/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	64.80		64.80								2,190.00	
DL	0.10		0.10								10.00	

Irrigation Water - Canal

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 09/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.14		2.80								117.00	78
DL	1.00		0.10								10.00	10

D. SOIL ANALYSES

Casavans

Casa

Sample and source description: Casa

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			37.90				
DL			2.00				

The 20

The 20

Sample and source description: The 20

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			115.00				
DL			2.00				

The 25

The 25

The 25

Sample and source description: The 25

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			208.00				
DL			2.00				

The 4

4 AC

Sample and source description: 4 AC

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.50				
DL			0.01				

E. PLANT TISSUE ANALYSES

Casavans - 11/05/2022: Oats, silage-soft dough

Casavans

Sample and source description: Casavans

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,100.00	5,000.00	45,000.00		10.00
DL	100.00	100.00	100.00		0.01

Casavans - 05/14/2023: Corn, silage

Casavans - 05/14/2023: Corn, silage

Casavans

Sample and source description: Casavans

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,500.00	7,900.00		4.40
DL	100.00	100.00	100.00		0.01

The 20 - 11/05/2022: Oats, silage-soft dough

The 20

Sample and source description: The 20

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	4,800.00	37,100.00		12.40
DL	100.00	100.00	100.00		0.01

The 20 - 05/14/2023: Corn, silage

The 20 + The 4

Sample and source description: The 20 + The 4

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	3,300.00	23,500.00		7.50
DL	100.00	100.00	100.00		0.01

The 25 - 11/05/2022: Oats, silage-soft dough

The 25

Sample and source description: The 25

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	4,800.00	41,300.00		10.20
DL	100.00	100.00	100.00		0.01

The 25 - 05/14/2023: Corn, silage

The 25

Sample and source description: The 25

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	3,300.00	20,700.00		7.70
DL	100.00	100.00	100.00		0.01

The 4 - 11/05/2022: Oats, silage-soft dough

The 4

Sample and source description: The 4

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	4,800.00	37,100.00		12.40
DL	100.00	100.00	100.00		0.01

The 4 - 05/14/2023: Corn, silage

The 20 + The 4

Sample and source description: The 20 + The 4

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	3,300.00	23,500.00		7.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

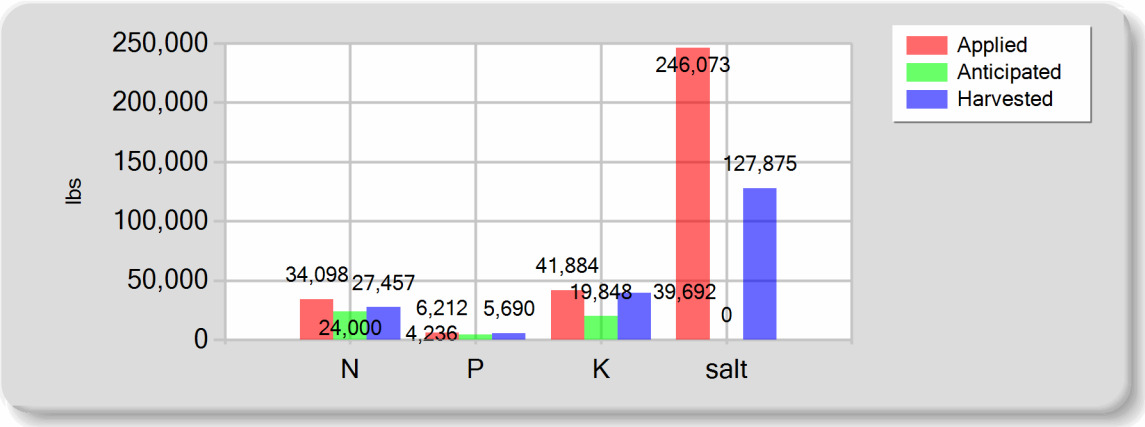
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

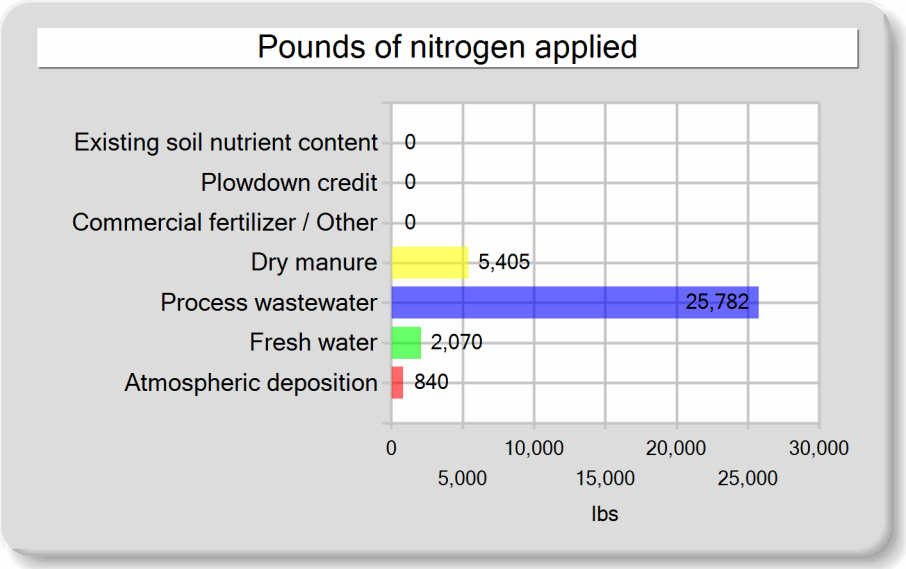
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

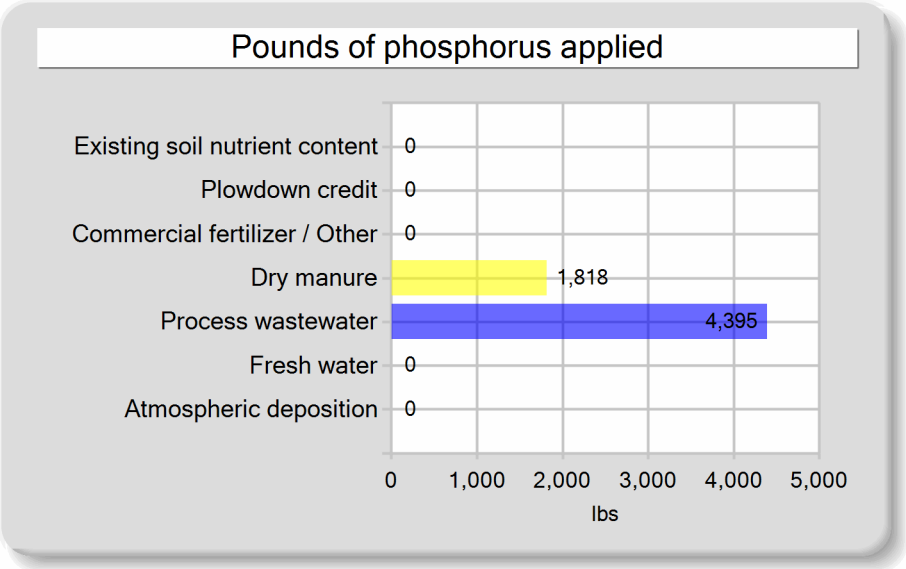
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	5,405.32	1,817.51	6,656.33	0.00
Process wastewater	25,782.12	4,394.54	35,228.00	194,649.63
Fresh water	2,070.10	0.00	0.00	51,422.90
Atmospheric deposition	840.00	0.00	0.00	0.00
Total nutrients applied	34,097.54	6,212.05	41,884.33	246,072.53
Anticipated crop nutrient removal	24,000.00	4,236.00	19,848.00	0.00
Actual crop nutrient removal	27,457.26	5,690.17	39,691.71	127,875.05
Nutrient balance	6,640.27	521.89	2,192.62	118,197.48
Applied to removed ratio	1.24	1.09	1.06	1.92

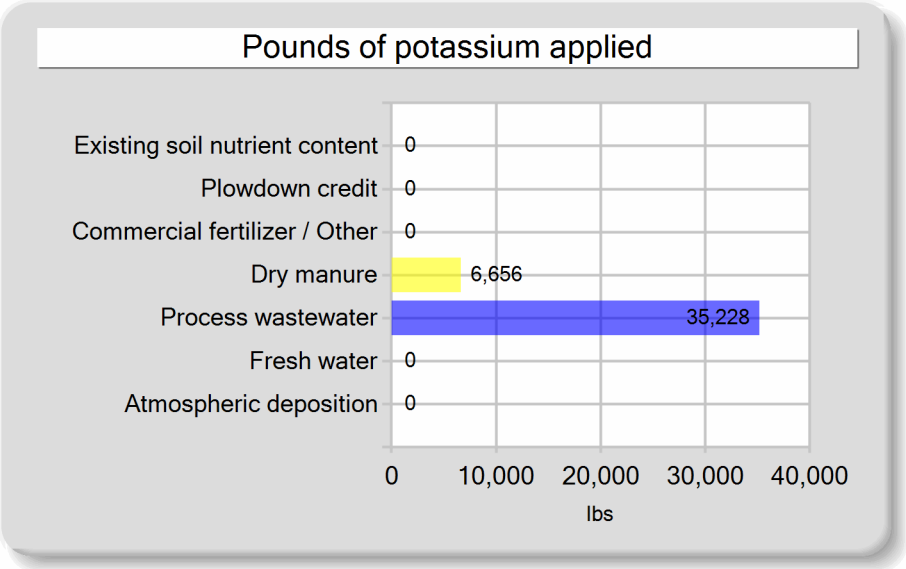
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

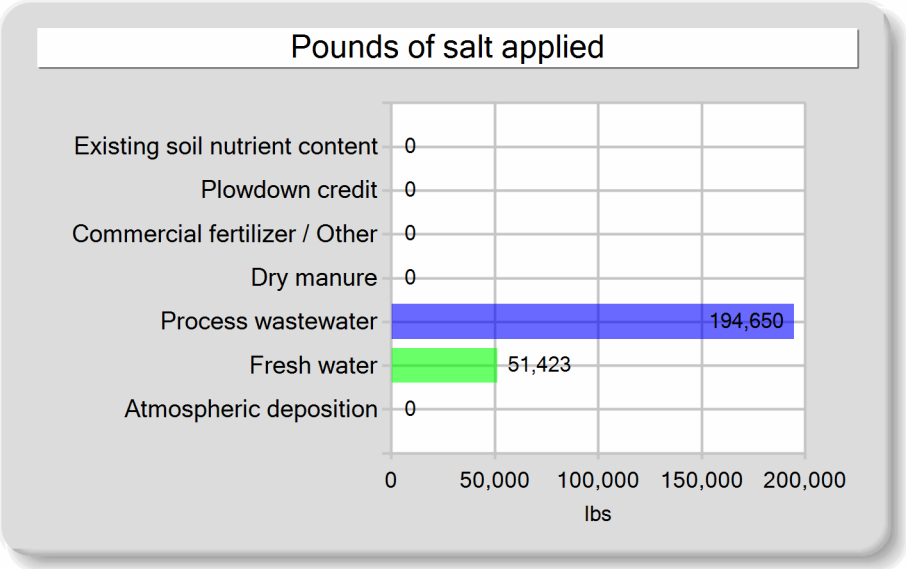


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Ken Van Foeken (Jun 21, 2024 10:26 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ken VanFoeken	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/21/2024	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator:	**Ken or Kurtis Van Foeken**
Name of Dairy Facility:	**Ken Van Foeken Dairy #2**
Facility Address:	**9735 Washington Road, Hilmar, CA 95324**
Contact Person Name and Phone Number:	**Kurtis Van Foeken (209) 658-1702**

Manure Hauler Information:

Name of Hauling Company/Person:	**Ken Van Foeken Dairy #2**
Address of Hauling Company /Person:	**9735 Washington Road, Hilmar, CA 95324**
Contact Person and Phone Number:	**Kurtis Van Foeken (209) 658-1702**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
B & D Dairy	**9579 Hultberg Road**	**Hilmar**	**95324**	**(209) 652-3534**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
Short Ave & Cummings Road	**Hilmar**	**95324**	**Merced County**

Dates Hauled: **3/3/23, 4/5/23, 6/16/23, 6/27/23, 7/5/23, & 7/14/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____Tons

Manure Solids Content (if amount reported in tons): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 5,992,390 Gallons

Method used to determine volume of process wastewater: Pump Run Time x Total Hours

Written Agreement:

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Ken Van Foeken (Jun 21, 2024 10:26 PDT) Date: 06/21/2024

Hauler's Signature: Ken Van Foeken (Jun 21, 2024 10:26 PDT) Date: 06/21/2024

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Ken or Kurtis Van Foeken**

Name of Dairy Facility: **Ken Van Foeken Dairy #2**

Facility Address: **9375 Washington Road, Hilmar, CA 95324**

Contact Person Name and Phone Number: **Kurtis Van Foeken (209) 658-1702**

Manure Hauler Information:

Name of Hauling Company/Person: **Silva & Sons, LLC**

Address of Hauling Company /Person: **23616 Williams Avenue, Hilmar, CA 95324**

Contact Person and Phone Number: **Frank Silva (209) 678-4027**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Broker** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Silva & Sons, LLC	23616 Williams Avenue	Hilmar	95324	(209) 678-4027

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
23616 Williams Avenue	Hilmar	95324	

Dates Hauled: **6/22/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **2,400** Tons

Manure Solids Content (if amount reported in tons): **84.3%**

Method used to determine amount of manure: **Number of Truckloads x Average Truck Weight**

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Ken Van Foeken (Jun 21, 2024 10:26 PDT) Date: 06/21/2024

Hauler's Signature: Date: 6/22/2023

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Ken Van Foeken Dairy #2

Received: 11/28/2023 13:50
Reported: 11/30/2023 09:20

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K1220-01	NE Well	Ag Water	T. Madruga		11/27/2023 17:36
23K1220-02	NW Well	Ag Water	T. Madruga		11/27/2023 17:30
23K1220-03	S Well	Ag Water	T. Madruga		11/27/2023 17:38

Default Cooler Temperature on Receipt °C: 2.1
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Ken Van Foeken Dairy #2

Received: 11/28/2023 13:50
Reported: 11/30/2023 09:20

Sample Results

Sample: NE Well
23K1220-01 (Water)

Sampled: 11/27/2023 17:36
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.74**	mmhos/cm	0.01	1		11/29/23 14:10	SM 2510 B		BEK0810
Electrical Conductivity umhos	**1740**	umhos/cm	10.0	1		11/29/23 14:10	SM 2510 B		BEK0810
Ammonia (as N)	ND	mg/L	0.00	1		11/27/23 17:36	Field		BEK0802
Nitrate Nitrogen as NO3N	**68.0**	mg/L	0.1	1	10	11/28/23 22:12	EPA 300.0		BEK0809
Temperature	**25.0**	units	0.0	1		11/29/23 14:10	SM 4500-H+	H	BEK0810
pH	**7.6**	units	1.0	1		11/29/23 14:10	SM 4500-H+	H	BEK0810

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Ken Van Foeken Dairy #2

Received: 11/28/2023 13:50
Reported: 11/30/2023 09:20

Sample Results
(Continued)

Sample: NW Well
23K1220-02 (Water)

Sampled: 11/27/2023 17:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.71**	mmhos/cm	0.01	1		11/29/23 14:12	SM 2510 B		BEK0810
Electrical Conductivity umhos	**1710**	umhos/cm	10.0	1		11/29/23 14:12	SM 2510 B		BEK0810
Ammonia (as N)	ND	mg/L	0.00	1		11/27/23 17:30	Field		BEK0802
Nitrate Nitrogen as NO3N	**63.7**	mg/L	0.1	1	10	11/28/23 22:33	EPA 300.0		BEK0809
Temperature	**25.0**	units	0.0	1		11/29/23 14:12	SM 4500-H+	H	BEK0810
pH	**7.6**	units	1.0	1		11/29/23 14:12	SM 4500-H+	H	BEK0810

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Ken Van Foeken Dairy #2

Received: 11/28/2023 13:50
Reported: 11/30/2023 09:20

Sample Results
(Continued)

Sample: S Well
23K1220-03 (Water)

Sampled: 11/27/2023 17:38
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.19**	mmhos/cm	0.01	1		11/29/23 14:13	SM 2510 B		BEK0810
Electrical Conductivity umhos	**2190**	umhos/cm	10.0	1		11/29/23 14:13	SM 2510 B		BEK0810
Ammonia (as N)	ND	mg/L	0.00	1		11/27/23 17:38	Field		BEK0802
Nitrate Nitrogen as NO3N	**64.8**	mg/L	0.1	1	10	11/29/23 01:16	EPA 300.0		BEK0809
Temperature	**25.0**	units	0.0	1		11/29/23 14:13	SM 4500-H+	H	BEK0810
pH	**7.6**	units	1.0	1		11/29/23 14:13	SM 4500-H+	H	BEK0810

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Received: 11/28/2023 13:50
Reported: 11/30/2023 09:20

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0809										
Blank (BEK0809-BLK1)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0809-BLK2)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0809-BLK3)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0809-BS1)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEK0809-BS2)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	5.5		0.1	mg/L	5.000		110	90-110		
Duplicate (BEK0809-DUP1)		**Source: 23K1216-02**			Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	4.1		0.1	mg/L		4.0			1.74	10
Duplicate (BEK0809-DUP2)		**Source: 23K1221-02**			Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L		4.9			0.123	10
Matrix Spike (BEK0809-MS1)		**Source: 23K1216-02**			Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	9.2		0.1	mg/L	5.000	4.0	104	90-110		
Matrix Spike (BEK0809-MS2)		**Source: 23K1221-02**			Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	10.2		0.1	mg/L	5.000	4.9	107	90-110		
Reference (BEK0809-SRM1)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0809-SRM2)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0809-SRM3)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Submitted By: Rui & John Avila
Ranch: Ken Van Foeken Dairy #2

Received: 11/28/2023 13:50
Reported: 11/30/2023 09:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0810										
Blank (BEK0810-BLK1)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.2		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0810-BLK2)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.8		1.0	units						
Blank (BEK0810-BLK3)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.8		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEK0810-DUP1)		**Source: 23K1215-02**			Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	1.83		0.01	mmhos/cm		1.85			0.766	10
Electrical Conductivity umhos	1830		10.0	umhos/cm		1850			0.766	10
pH	7.8		1.0	units		7.8			0.128	10
Duplicate (BEK0810-DUP2)		**Source: 23K1221-02**			Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	0.46		0.01	mmhos/cm		0.46			0.810	10
Electrical Conductivity umhos	455		10.0	umhos/cm		459			0.810	10
pH	8.0		1.0	units		8.0			0.00	10
Reference (BEK0810-SRM1)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	565			umhos/cm	538.0		105	90-110		
Reference (BEK0810-SRM2)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	ND		0.01	mmhos/cm				90-110		
Reference (BEK0810-SRM3)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		
Reference (BEK0810-SRM4)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
pH	6.6		1.0	units				97.5-102.5		
Reference (BEK0810-SRM5)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
pH	6.5		1.0	units				97.5-102.5		
Reference (BEK0810-SRM6)					Prepared: 11/28/2023	Analyzed: 11/29/2023				

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Received: 11/28/2023 13:50
Reported: 11/30/2023 09:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0810 (Continued)										
Reference (BEK0810-SRM6)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0810-SRM7)					Prepared: 11/28/2023	Analyzed: 11/29/2023				
pH	4.0			units	4.000		100	67021-101.3		

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11/28/23 13:50 23K1220

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

JG

No. Samples 3 No. of Bottles: 3

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Ken Van Foeken Dairy #2
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	NE Well	11/27/2023	1736	2.1	ND
2	NW Well	11/27/2023	1730	2.3	ND
3	S Well	11/27/2023	1738	2.1	ND
4					
5					
6					
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/27/2023 @ 1738	11/28/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	11/28/2023 @ 900	11-28-23 9:00:20
Third				
Fourth	JLF	DLI	11/28/23 13:50	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay, in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

Invoicing Information:		*Shipping*	
Sampling hrs		$	In
Miles		$	Out
Consulting			
Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/28/23 13:50 23K1220

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										